Initial Design & ADS Simulation Analysis for 5.8 GHz Branch-Line Coupler (BLC)

1 Introduction & Substrate Specifications

A microstrip Branch-Line Quadrature Hybrid Coupler (BLC) was designed and optimized in Keysight ADS for center frequency $f_0 = 5.8$ GHz. The design uses a low-loss substrate (`MSub1`) to achieve equal -3 dB power division at Ports 2 and 3 with a quadrature (90°) output phase difference.

Table 1: Substrate (`MSub1`) and Microstrip Optimization Parameters (`VAR1`)

Substrate Parameter	**Value**	**Optimized Variable (`VAR1`)**	**Value**
Substrate Height (H)	0.51 mm	Horizontal Line Width (w_h)	1.91414 mm ($Z_0/\sqrt{2} = 35.35\,\Omega$)
Relative Permittivity (ε_r)	2.2	Horizontal Line Length (l_h)	8.29545 mm ($\lambda/4$)
Conductor Thickness (T) / Cond.	0.035 mm / 5.8×10^7 S/m	Vertical Line Width (w_v)	1.12626 mm ($Z_0 = 50.0\,\Omega$)
Loss Tangent ($\tan\delta$) / Roughness	0.0009 / 0.001 mm	Vertical Line Length (l_v)	9.25253 mm ($\lambda/4$)

2 Design Equations & Optimization Goals

The BLC uses quarter-wave ($\lambda/4$) microstrip branches. Horizontal branches require $Z_{0h} = 35.35\,\Omega$, and vertical branches require $Z_{0v} = 50\,\Omega$. Optimization goals in ADS (`Optim1`) were set across $5.0 - 6.5$ GHz:

$$\text{Goal 1: } \mathrm{dB}(S_{4,1}) \to \min, \quad \text{Goal 2: } \mathrm{dB}(S_{1,1}) \to \min, \quad \text{Goal 3: } \mathrm{dB}(S_{2,1}) - \mathrm{dB}(S_{3,1}) \to 0 \text{ dB}, \quad \text{Goal 4: } \angle S_{2,1} - \angle S_{3,1} \to 90^\circ \tag{1}$$

3 ADS Schematic Capture & Board Layout

The circuit configuration incorporates microstrip line elements (`MLIN`), microstrip tee junctions (`MTEE_ADS`), $50\,\Omega$ port terminations (`TermG1--TermG4`), and S-parameter simulation parameters (`SP1`: 5.0 to 6.5 GHz).

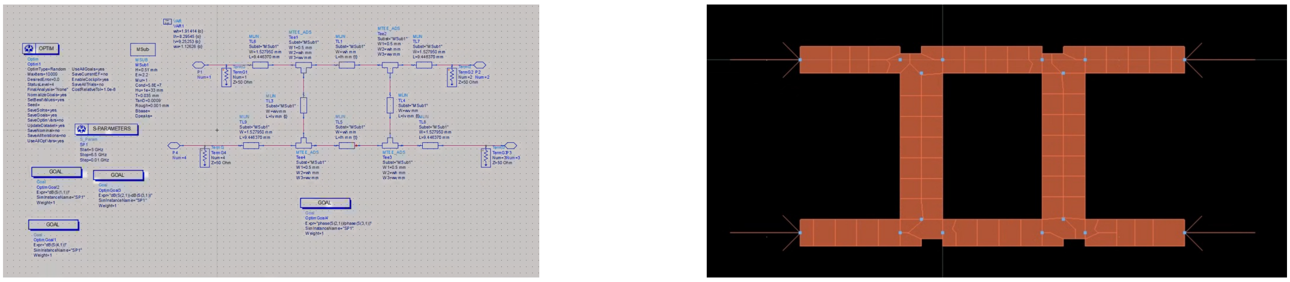

Figure 1: Keysight ADS schematic capture and generated 2D microstrip board layout of the 5.8 GHz BLC.

4 ADS Simulation Results

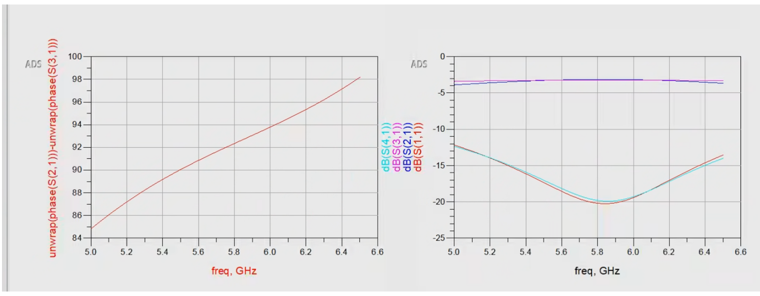

Figure 2: Simulated Phase Difference $\angle S_{2,1} - \angle S_{3,1}$ (left) and S-parameters $S_{11}, S_{21}, S_{31}, S_{41}$ (right).

4.1 Performance Summary at 5.8 GHz

Table 2: ADS S-Parameter Simulation Performance at Center Frequency $f_0 = 5.8$ GHz

Parameter	**Coupling ($S_{3,1}$)**	**Thru ($S_{2,1}$)**	**Return Loss ($S_{1,1}$)**	**Isolation ($S_{4,1}$)**	**Phase Balance ($\Delta\phi$)**
ADS Simulation	-3.1 dB	-3.1 dB	-20.2 dB	-20.1 dB	92.4° ($\approx 90°$ quadrature)